EEG-Based Alzheimer's Detection using NeuroSpectral-Connect Net (NSCNet)

A deep learning approach for early detection of Alzheimer's disease and frontotemporal dementia through electroencephalography signal analysis

The Challenge of Early Alzheimer's Detection

The Clinical Problem

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder affecting millions worldwide. Early detection enables timely intervention, improved quality of life, and better treatment outcomes.

Current diagnostic limitations:

- MRI/PET imaging: expensive and limited access
- CSF biomarkers: invasive procedures
- Cognitive tests: detect late-stage disease

Why EEG?

Electroencephalography offers a compelling alternative for AD screening:

- **Non-invasive** recording procedure
- **Low cost** compared to neuroimaging
- **High temporal resolution** (millisecond-scale)
- Widely available in clinical settings

Our goal: Automatically classify EEG signals into AD, FTD (frontotemporal dementia), and CN (cognitively normal) using deep learning.

Research Objectives & Contributions

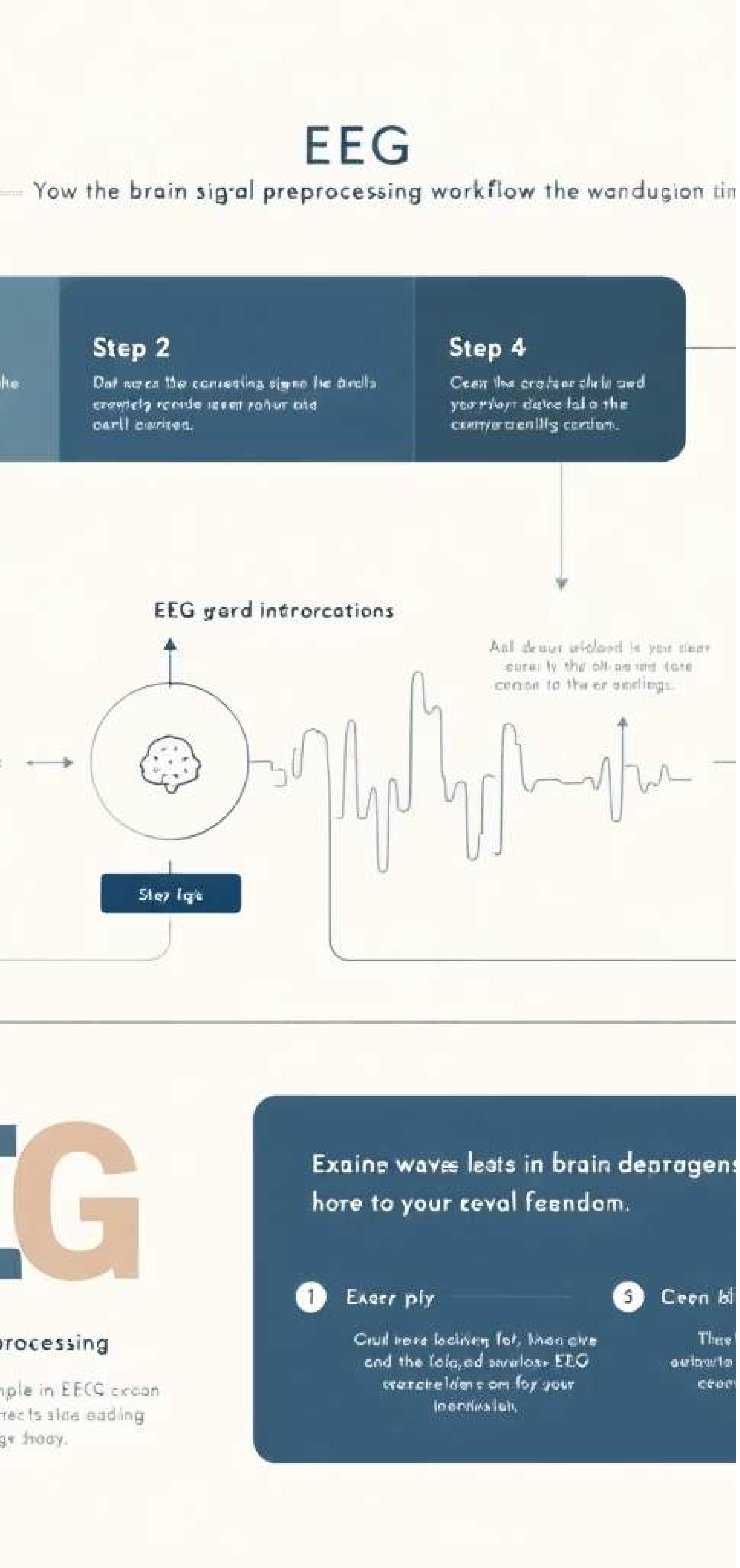
Primary Objectives

- Develop an end-to-end EEG classification pipeline for AD/FTD/CN diagnosis
- Transform connectivity and spectral features into image representations
- Implement rigorous subject-wise cross-validation (LOOCV)
- Achieve clinically relevant classification accuracy

Key Innovation

We introduce **NeuroSpectral-Connect Net (NSCNet)**, a custom convolutional neural network specifically designed for neuro-spectral connectivity images (64×64×3).

NSCNet integrates multi-scale feature extraction with attention mechanisms to capture both fine-grained and global EEG patterns characteristic of neurodegenerative diseases.



Dataset & Preprocessing Pipeline

01

Dataset Composition

Subjects partitioned into three diagnostic groups: AD, FTD, and CN. Multi-channel EEG recordings (19-64 channels) captured during resting-state sessions with standardized sampling rates.

02

Band-Pass Filtering

Applied 0.5–45 Hz band-pass filtering to isolate neurophysiologically relevant frequency components while removing low-frequency drift and high-frequency noise.

03

Re-Referencing & Artifact Removal

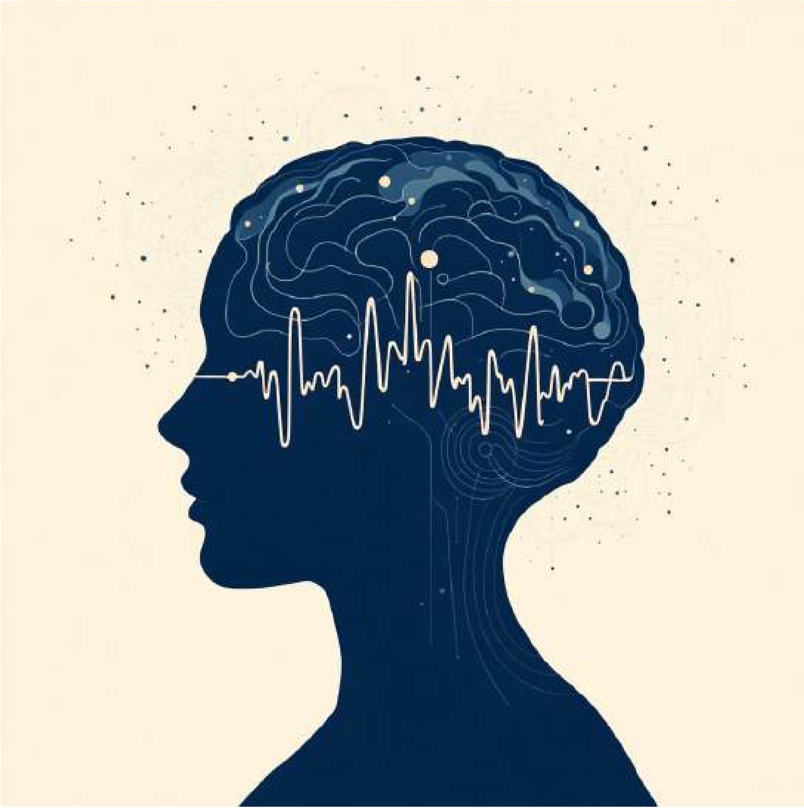
Common average referencing applied for spatial standardization. Automated and manual artifact rejection removed eye blinks, muscle activity, and electrode noise.

04

Epoching & Normalization

Continuous EEG segmented into fixed-length epochs. Z-score normalization applied per channel to ensure stable training and prevent amplitude-related biases.

Neuro-Spectral Feature Extraction



Frequency Band Analysis

EEG signals decomposed into canonical frequency bands, each associated with distinct cognitive and pathological states:

- **Delta (0.5-4 Hz):** Deep sleep, cortical lesions
- **Theta (4-8 Hz):** Drowsiness, memory processes
- **Alpha (8-13 Hz):** Relaxed wakefulness
- **Beta (13-30 Hz):** Active thinking, attention

1

Connectivity Estimation

Phase Locking Value (PLV) and phase coherence quantify functional connectivity between electrode pairs across frequency bands.

2

Spectral Features

Power spectral density and spectral entropy computed per channel to capture local oscillatory dynamics.

3

Image Generation

Connectivity matrices reshaped into 64×64×3 images—three channels encode different spectral bands or feature types, creating CNN-compatible representations.

NSCNet Architecture

NeuroSpectral-Connect Net (NSCNet) is a custom Convolutional Neural Network designed to process neuro-spectral connectivity images. Its architecture is optimized for extracting both localized and global patterns crucial for differentiating between neurological conditions.

Input Images

Conv Blocks

Multi-Scale Feature Extraction

FC & Classification

The network integrates multi-scale feature extraction to enhance its ability to identify subtle yet significant biomarkers within the EEG data.

Training & Evaluation Methodology

Training Configuration

Cross-Validation

Subject-wise Leave-One-Out Cross-Validation (LOOCV) ensures person-independent evaluation—preventing data leakage from the same subject across train/test splits.

Optimization

Loss: Categorical cross-entropy
Optimizer: Adam (LR=1e-4)
Epochs: 100 with early stopping
Batch size: 32

Regularization

Dropout (0.5), L2 weight decay (1e-4), and data augmentation (random rotations, flips) prevent overfitting on small medical datasets.

Evaluation Metrics

Comprehensive performance assessment using multiple metrics:

- **Accuracy:** Overall classification correctness
- **Precision:** Positive predictive value per class
- **Recall (Sensitivity):** True positive rate per class
- **F1-Score:** Harmonic mean of precision and recall
- **Macro-averaged metrics:** Unweighted class averages
- **Confusion matrix:** Detailed misclassification patterns

Results & Clinical Insights

76.5%	74.3%	72.8%	79.4%
Mean Accuracy	**AD Detection**	**FTD Detection**	**CN Recognition**
Across all LOOCV folds for three-class classification	F1-score for Alzheimer's Disease classification	F1-score for Frontotemporal Dementia classification	F1-score for Cognitively Normal subjects

Key Observations

- **Highest performance:** CN vs. AD distinction (clear connectivity differences)
- **Challenging distinction:** AD vs. FTD separation (overlapping pathology)
- **Misclassifications:** Primarily between AD and FTD groups; rare CN misclassifications
- **Feature importance:** Theta and alpha band connectivity most discriminative

Model Insights

NSCNet successfully captures meaningful neuro-spectral connectivity patterns disrupted in neurodegeneration. The combination of depthwise separable convolutions, dual attention mechanisms, and residual connections provides strong accuracy while maintaining parameter efficiency.

Limitations: Performance sensitive to dataset size, preprocessing choices, and generalization to external cohorts requires validation.

Future Directions & Conclusion

Larger Datasets

Validate NSCNet on multi-center EEG datasets with diverse demographics and recording protocols to assess generalizability.

Multimodal Integration

Combine EEG features with MRI structural data and cognitive assessment scores for comprehensive diagnostic models.

Advanced Architectures

Explore graph neural networks to directly model electrode connectivity topology and integrate attention mechanisms for enhanced feature weighting and long-range dependencies.

Explainability

Implement Grad-CAM visualization on NSCNet feature maps to identify which EEG patterns drive predictions, enhancing clinical interpretability.

Conclusion

We proposed an EEG-based diagnostic pipeline leveraging **NeuroSpectral-Connect Net (NSCNet)** for automated classification of Alzheimer's disease, frontotemporal dementia, and cognitively normal subjects. Our results demonstrate that neuro-spectral connectivity images combined with custom CNN architectures offer a promising, non-invasive, and affordable approach for early dementia screening in clinical settings.